1. Problem Statement

Threaded BST Insertion: Write a C++ program to insert a new node into an inorder-threaded binary search tree (BST) and update the affected predecessor and successor threads.

2. Objectives

- **Eliminate traversal overhead:** avoid runtime stacks and recursion during BST traversal by leveraging thread connections.
- **Efficient memory utilization:** repurpose the unused NULL child pointers of leaf and partially filled nodes as directional threads that reference the immediate inorder predecessor or successor.
- **Maintain structural integrity:** link the structural parent and child nodes correctly while adjusting the neighbouring predecessor and successor threads on every insertion.
- **Achieve linear traversal complexity:** implement a deterministic O(n) iterative inorder loop that visits the nodes strictly through thread addresses.